

22AIE111 OBJECT ORIENTED PROGRAMMING IN JAVA

S2 B.Tech CSE-AI

Department of Computer Science and Engineering

Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – I (Analysis)

Problem Identification & Class Mapping (CO1)

Group No : 2

Team Members :

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1. Problem Title : **Online Food Ordering System**

2. Problem Description :

Traditional ordering methods lead to order confusion, billing errors, and delays.

3. Identified Entities (Classes)

Class Name	Description
Customer	Represents a user who orders food
Restaurant	Handles food preparation and menu management
Menu	Contains list of food items available

Order	Represents a food order placed by customer
Payment	Handles payment transactions
DeliveryAgent	Responsible for delivering orders
OrderManagementSystem	Core system managing all operations

4. Class Attributes and Behaviours

Class Name	Attributes	Methods(behaviours)
Customer	customerId name address phoneNumber	browseMenu() placeOrder() makePayment() trackOrder()
Restaurant	restaurantId name location menuList	receiveOrder() prepareFood() updateMenu()
Menu	menuId itemList prices	displayMenu() updateItem()
Order	orderId items quantity totalAmount orderStatus	createOrder() updateStatus() calculateTotal()
DeliveryAgent	agentId name contactNumber	pickOrder() deliverOrder() updateDeliveryStatus()

	availabilityStatus	
OrderManagementSystem	systemId ordersList customersList	trackOrders() assignDeliveryAgent() updateOrderStatus()
Payment	paymentId amount paymentMethod paymentStatus	processPayment() verifyPayment()